

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Lanthanum(III) 2-methoxyethoxide, 99.9% (REO), 5% w/v in 2-methoxyethanol

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>42522</b>                                                                                         |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 3

#### Health hazards

Acute Oral Toxicity  
Acute Dermal Toxicity  
Acute Inhalation Toxicity - Vapors  
Reproductive Toxicity  
Specific target organ toxicity - (single exposure)  
Specific target organ toxicity - (repeated exposure)

Category 4  
Category 4  
Category 4  
Category 1B  
Category 1  
Category 2

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Exclamation Mark



Health Hazard

**Signal Word**

**Danger**

**Hazard Statements**

H226 - Flammable liquid and vapor

H360 - May damage fertility or the unborn child

H370 - Causes damage to organs

H373 - May cause damage to organs through prolonged or repeated exposure

H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled

**Precautionary Statements**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P240 - Ground and bond container and receiving equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P312 - Call a POISON CENTER or doctor if you feel unwell

P330 - Rinse mouth

P363 - Wash contaminated clothing before reuse

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P308 + P311 - IF exposed or concerned: Call a POISON CENTER or doctor

P403 + P235 - Store in a well-ventilated place. Keep cool

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component                        | CAS No      | Weight % |
|----------------------------------|-------------|----------|
| 2-Methoxyethanol                 | 109-86-4    | 95.00    |
| Lanthanum(III) 2-methoxyethoxide | 115685-56-8 | 5.00     |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

|                                     |                                                                                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Ingestion                           | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| Skin Contact                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| Eye Contact                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| General Advice                      | If symptoms persist, call a physician.                                                                                                           |
| Self-Protection of the First Aider  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| First Aid Facilities                | Eyewash, safety shower and washroom.                                                                                                             |
| Most important symptoms and effects | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting                                     |
| Notes to Physician                  | Treat symptomatically.                                                                                                                           |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Lanthanum oxide.

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

#### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

#### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

#### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component        | Australia                               | New Zealand WEL                                    | ACGIH TLV            | The United Kingdom                                                                                                 | Germany                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|-----------------------------------------|----------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-Methoxyethanol | TWA: 5 ppm<br>TWA: 16 mg/m <sup>3</sup> | TWA: 0.1 ppm<br>TWA: 0.3 mg/m <sup>3</sup><br>Skin | TWA: 0.1 ppm<br>Skin | STEL: 3 ppm 15 min<br>STEL: 9 mg/m <sup>3</sup> 15 min<br>TWA: 1 ppm 8 hr<br>TWA: 3 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 1 ppm (8 Stunden). AGW - exposure factor 8<br>TWA: 3.2 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 1 ppm (8 Stunden). MAK applies for the sum of the concentrations of 2-Methoxyethanol and its Acetate in air<br>TWA: 3.2 mg/m <sup>3</sup> (8 Stunden). MAK applies for the sum of the concentrations of 2-Methoxyethanol and its Acetate in air<br>Höhepunkt: 8 ppm<br>Höhepunkt: 25.6 mg/m <sup>3</sup><br>Haut |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

| Component        | Australia | New Zealand | European Union | United Kingdom | Germany                                                      |
|------------------|-----------|-------------|----------------|----------------|--------------------------------------------------------------|
| 2-Methoxyethanol |           |             |                |                | Methoxyacetic acid: 15 mg/g Creatinine urine (end of shift ) |

## Exposure Controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Butyl rubber   | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Skin and body protection

Long sleeved clothing

#### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

#### Recommended Filter type:

Multi-purpose/ABEK conforming to EN14387 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

### Environmental exposure controls

No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

#### Appearance

##### Physical State

Liquid

#### Odor

Characteristic

#### Odor Threshold

No data available

#### pH

4

#### Melting Point/Range

No data available

#### Softening Point

No data available

#### Boiling Point/Range

124 °C / 255.2 °F

#### Flash Point

40 °C / 104 °F

**Method -** No information available

#### Evaporation Rate

No data available

#### Flammability (solid,gas)

Not applicable

Liquid

#### Explosion Limits

No data available

#### Vapor Pressure

23 hPa @ 20 °C

#### Vapor Density

No data available

(Air = 1.0)

#### Specific Gravity / Density

0.96 g/cm3

@ 20 °C

#### Bulk Density

Not applicable

Liquid

|                                         |                          |                                        |
|-----------------------------------------|--------------------------|----------------------------------------|
| Water Solubility                        | Immiscible               |                                        |
| Solubility in other solvents            | No information available |                                        |
| Partition Coefficient (n-octanol/water) |                          |                                        |
| Component                               | log Pow                  |                                        |
| 2-Methoxyethanol                        | -0.77                    |                                        |
| Autoignition Temperature                | No data available        |                                        |
| Decomposition Temperature               | No data available        |                                        |
| Viscosity                               | No data available        |                                        |
| Explosive Properties                    |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties                    | No information available |                                        |
| <b>Other information</b>                |                          |                                        |
| Molecular Formula                       | C9 H21 LaO6              |                                        |
| Molecular Weight                        | 364.17                   |                                        |

## Section 10 - Stability and Reactivity

|                                  |                                                                           |
|----------------------------------|---------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                |
| Stability                        | Moisture sensitive.                                                       |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition.         |
| Incompatible Materials           | None known.                                                               |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Lanthanum oxide. |
| Hazardous Polymerization         | No information available.                                                 |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                     |            |
|---------------------|------------|
| (a) acute toxicity; |            |
| Oral                | Category 4 |
| Dermal              | Category 4 |
| Inhalation          | Category 4 |

#### Toxicology data for the components

| Component        | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation             |
|------------------|---------------------------|------------------------------|-----------------------------|
| 2-Methoxyethanol | LD50 = 2370 mg/kg ( Rat ) | LD50 = 1280 mg/kg ( Rabbit ) | LC50 = 1478 ppm ( Rat ) 7 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| (f) carcinogenicity;        | No data available<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;  | Category 1B                                                                    |
| (h) STOT-single exposure;   | Category 1                                                                     |
| Results / Target organs     | Immune system                                                                  |
| (i) STOT-repeated exposure; | Category 2                                                                     |
| Target Organs               | Thymus.                                                                        |
| (j) aspiration hazard;      | No data available                                                              |

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

**Ecotoxicity effects** May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component        | Freshwater Fish                                                                                                                                                     | Water Flea | Freshwater Algae | Microtox |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|----------|
| 2-Methoxyethanol | LC50: = 9650 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 16000 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 10000 mg/L, 96h static (Lepomis macrochirus) |            |                  |          |

**Persistence and Degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
**Persistence** May persist, based on information available.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.  
**Bioaccumulative Potential** May have some potential to bioaccumulate

| Component        | log Pow | Bioconcentration factor (BCF) |
|------------------|---------|-------------------------------|
| 2-Methoxyethanol | -0.77   | No data available             |

**Mobility** Spillage unlikely to penetrate soil. The product is insoluble and floats on water. Is not likely mobile in the environment due to its low water solubility  
**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

## Section 14 - Transport Information

**IMDG/IMO**

UN-No UN1188  
Proper Shipping Name ETHYLENE GLYCOL MONOMETHYL ETHER  
Hazard Class 3  
Packing Group III

**ADG**

UN-No UN1188  
Proper Shipping Name ETHYLENE GLYCOL MONOMETHYL ETHER  
Hazard Class 3  
Packing Group III

| Component                              | Hazchem Code |
|----------------------------------------|--------------|
| 2-Methoxyethanol<br>109-86-4 ( 95.00 ) | 2Y           |

**IATA**

UN-No UN1188  
Proper Shipping Name ETHYLENE GLYCOL MONOMETHYL ETHER  
Hazard Class 3  
Packing Group III

Environmental hazards No hazards identified

Special Precautions No special precautions required

Additional information None known

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National Regulations Australia**

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                   | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                           |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-Methoxyethanol - 109-86-4 | Schedule 6 listed - except when separately specified in these Schedules, or in preparations containing<br>≤10% of such substances<br>Schedule 7 listed |

**Australian Industrial Chemicals Introduction Scheme (AICIS)**



| Component                   | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-----------------------------|-------------------------------------------------------------|------------------------|
| 2-Methoxyethanol - 109-86-4 | Present                                                     | -                      |

#### Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

#### National pollutant inventory Subject to reporting requirements

| Component                   | National pollutant inventory      |
|-----------------------------|-----------------------------------|
| 2-Methoxyethanol - 109-86-4 | 10 tonne/yr. Threshold category 1 |

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### International Inventories

| Component        | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| 2-Methoxyethanol | X    | X     | 203-713-7 | -      | X    | X   | -    | X     | X    | X    | X     | KE-23272 |

**Legend:** X - Listed, '-' - Not Listed, S - Indicates a substance that is identified in a proposed or final Significant New Use Rule. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Not applicable.

| Component                        | CAS No      | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------------------------|-------------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 2-Methoxyethanol                 | 109-86-4    | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Lanthanum(III) 2-methoxyethoxide | 115685-56-8 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component        | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 2-Methoxyethanol | -                                                                   | Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - 203-713-7 - Toxic for reproduction, Article 57c                                 |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### **Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

### **Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Revision Date** 19-Nov-2022

**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

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**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**